

```
$ docker run\
--volume=/:/rootfs:ro\
--volume=/var/run:/var/run:rw \
--volume=/sys:/sys:ro \
--volume=/var/lib/docker/:/var/lib/\
docker:ro \
--publish=8080:8080 \ --detach=true \
--name=cadvisor \
google/cadvisor:latest
```

cAdvisor can now be accessed at <http://localhost:8080>, as shown in Figure 3.

Prometheus

Prometheus is another GUI based open source tool for monitoring Docker containers and was developed by SoundCloud. Along with Docker Stats and cAdvisor, Prometheus also provides an alert mechanism based on some applied rules. It makes use of the exporter API to capture and store container metrics. The installation of this container monitoring tool can be done by executing the following commands.

Step 1: Download the latest release of Prometheus, and then extract it:

```
$ tar xvfz prometheus-*.tar.gz
$ cd prometheus-*
```

Step 2: Change to the directory containing the source code of Prometheus and run the following command, as depicted in Figure 4:

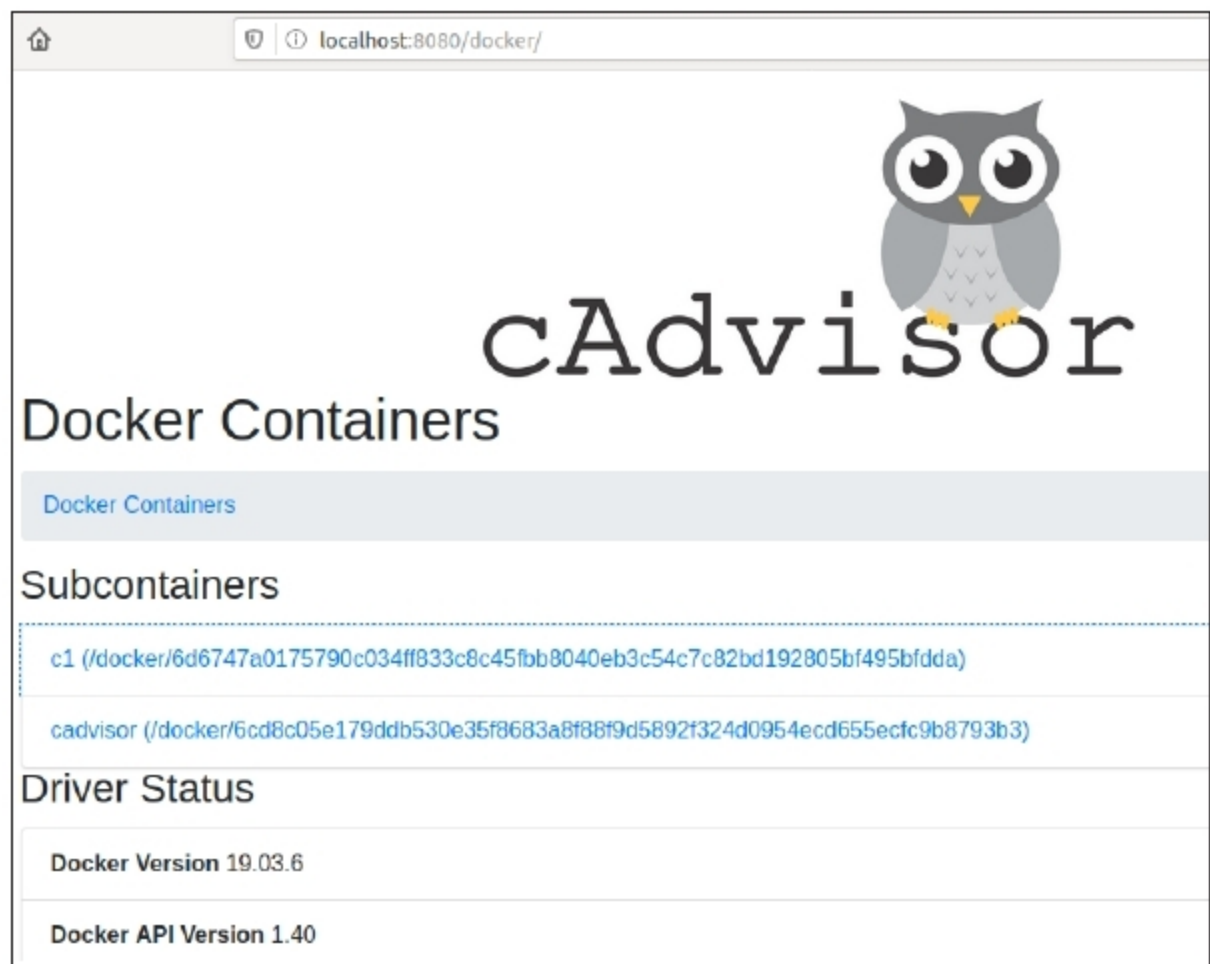


Figure 3: Dashboard for the cAdvisor monitoring tool

```
$ ./prometheus - -config.file =
prometheus.yml
```

Prometheus can be accessed at <http://localhost:9090>, as shown in Figure 5.

Sensu

Sensu is a container monitoring tool that provides support for all three container technologies, namely LXC, Docker, and RKT. Sensu is self-hosted and offers centralised metrics services. However, in the production environment, the deployment of this tool depends upon

various complementary services such as Sensu API and Sensu Core, as shown in Figure 6. The Sensu monitoring tool can be installed using the following commands.

Step 1: Install the Sensu backend, as follows. Add the Sensu repository:

```
curl -s https://packagecloud.io/
install/repositories/sensu/stable/
script.deb.sh | sudo bash
```

Next, install the *sensu-go-backend* package:

```
sudo apt-get install sensu-go-backend
```

```
root@server3:~/Downloads/prometheus-2.16.0.linux-amd64# ./prometheus --config.file=prometheus.yml
level=info ts=2020-02-15T12:14:39.587Z caller=main.go:295 msg="no time or size retention was set so using the default time retention" duration=15d
level=info ts=2020-02-15T12:14:39.588Z caller=main.go:331 msg="Starting Prometheus" version="(version=2.16.0, branch=HEAD, revision=b90be6f3233c03163d700e1452b54454ddce0ec)"
level=info ts=2020-02-15T12:14:39.588Z caller=main.go:332 build_context="(go=go1.13.8, user=root@7ea0ae865f12, date=20200213-23:50:02)"
level=info ts=2020-02-15T12:14:39.588Z caller=main.go:333 host_details="(Linux 4.15.0-47-generic #50-Ubuntu SMP Wed Mar 13 10:44:52 UTC 2019 x86_64 server3 (none))"
level=info ts=2020-02-15T12:14:39.588Z caller=main.go:334 fd_limits="(soft=1024, hard=4096)"
level=info ts=2020-02-15T12:14:39.588Z caller=main.go:335 vm_limits="(soft=unlimited, hard=unlimited)"
level=info ts=2020-02-15T12:14:39.684Z caller=web.go:508 component=web msg="Start listening for connections" address=0.0.0.0:9090
level=info ts=2020-02-15T12:14:39.684Z caller=main.go:661 msg="Starting TSDB ..."
level=info ts=2020-02-15T12:14:39.745Z caller=head.go:577 component=tsdb msg="replaying WAL, this may take awhile"
level=info ts=2020-02-15T12:14:39.846Z caller=head.go:625 component=tsdb msg="WAL segment loaded" segment=0 maxSegment=1
level=info ts=2020-02-15T12:14:39.846Z caller=head.go:625 component=tsdb msg="WAL segment loaded" segment=1 maxSegment=1
level=info ts=2020-02-15T12:14:39.849Z caller=main.go:676 fs_type=EXT4_SUPER_MAGIC
level=info ts=2020-02-15T12:14:39.849Z caller=main.go:677 msg="TSDB started"
level=info ts=2020-02-15T12:14:39.849Z caller=main.go:747 msg="Loading configuration file" filename=prometheus.yml
level=info ts=2020-02-15T12:14:39.918Z caller=main.go:775 msg="Completed loading of configuration file" filename=prometheus.yml
level=info ts=2020-02-15T12:14:39.919Z caller=main.go:630 msg="Server is ready to receive web requests."
```

Figure 4: Installing the Prometheus container monitoring tool